

Axcel®
CEFUROXIME-125
FOR ORAL SUSPENSION
(125mg/5ml)

COMPOSITION:

Each 5ml contains
 Cefuroxime Axetil
 equivalent to Cefuroxime 125mg

PRESENTATION:

Off white to pale yellow granules, which gives off white to pale yellow suspension with fruity flavor after reconstituted.

INDICATIONS:

Upper respiratory tract infections (for example: ear, nose and throat infections, such as otitis media, sinusitis, tonsillitis and pharyngitis). Lower respiratory tract infections (for example: pneumonia, acute bronchitis, and acute exacerbations of chronic bronchitis). Genito-urinary tract infections (for example: pyelonephritis, cystitis and urethritis). Gonorrhoea, acute uncomplicated gonococcal urethritis, and cervicitis. Skin and soft tissue infections (for example: furunculosis, pyoderma and impetigo). Treatment of early Lyme disease and subsequent prevention of late Lyme disease in adults and children over 12 years old.

PHARMACOLOGY:

Mechanism of Action:

Cefuroxime axetil owes its in vivo bactericidal activity to the parent compound cefuroxime. Cefuroxime axetil is an oral prodrug of the bactericidal cephalosporin antibiotic cefuroxime, which is resistant to most β - lactamases and is active against a wide range of Gram-positive and Gram-negative organisms. It is indicated for the treatment of infections caused by susceptible bacteria. Cefuroxime is a well characterized and effective antibacterial agent which has bactericidal activity against a wide range of common pathogens, including β -lactamase producing strains. Cefuroxime has good stability to bacterial β -lactamase, and consequently is active against many ampicillin-resistant or amoxycillin-resistant strains. The bactericidal action of cefuroxime results from inhibition of cell wall synthesis by binding to essential target proteins. Cefuroxime is usually active against the following organisms *in vitro* : -

Aerobes Gram - negative :

Haemophilus influenza (including ampicillin – resistant strains), *Haemophilus parainfluenzae*, *Moraxella* (*Branhamella*) *catarrhalis*, *Neisseria*

gonorrhoeae (including penicillinase and non - penicillinase producing strains), *Escherichia coli*, *Klebsiella spp.*, *Proteus mirabilis*, *Providencia spp.*, *Proteus reitteri*.

Aerobes Gram - positive :

Staphylococcus aureus (including penicillinase producing strains but excluding methicillin resistant strains), *Staphylococcus epidermidis* (including penicillinase producing strains but excluding methicillin resistant strains), *Streptococcus pyogenes* (and other β -haemolytic streptococci), *Streptococcus pneumoniae*, *Streptococcus Group B (Streptococcus agalactiae)*.

Anaerobes :

Gram-positive cocci and Gram-negative cocci (including *Peptococcus* and *Peptostreptococcus* species), Gram-positive bacilli (including *Clostridium* species), Gram-negative bacilli (including *Bacteroides* and *Fusobacterium* species) *Propionibacterium* spp.

Other organisms :

Borrelia burgdorferi.

The following organisms are not susceptible to Cefuroxime :

Clostridium difficile, *Pseudomonas spp.*, *Campylobacter spp.*, *Acinetobacter calcoaceticus*, *Listeria monocytogenes*, Methicillin resistant strains of *Staphylococcus aureus* and *Staphylococcus epidermidis* and *Legionella* spp.

Some strains of the following genera are not susceptible to Cefuroxime :

Enterococcus (*Streptococcus*) *faecalis*, *Morganella morganii*, *Proteus vulgaris*, *Enterobacter* spp., *Citrobacter* spp., *Serratia* spp. and *Bacteroides fragilis*.

Pharmacokinetics :

After oral administration cefuroxime axetil is absorbed from the gastrointestinal tract and rapidly hydrolysed in the intestinal mucosa and blood to release cefuroxime into the circulation. Absorption of cefuroxime axetil suspension is enhanced in the presence of food. The rate of absorption of cefuroxime from the suspension compared with the tablets is reduced, leading to later, lower peak serum levels and reduced systemic bioavailability (4-17% less). The serum half-life is between 1 and 1.5 hours. Protein binding has been variously stated as 33-50% depending on the methodology used. Cefuroxime is not metabolized and is excreted by glomerular filtration and tubular secretion. Concurrent administration of probenecid increases the area under the mean serum concentrations time curve by 50%. Serum levels of cefuroxime are reduced by dialysis.

DOSAGE AND ADMINISTRATION:

For oral administration only.

The usual course of therapy is seven days, (range 5 – 10 days).

For optimal absorption cefuroxime axetil should be taken after food.

Adults :

Most Infections : 250mg twice daily

Urinary tract infections : 125mg twice daily

Mild to moderate lower respiratory tract infections e.g. bronchitis :

250mg twice daily

More severe lower respiratory tract infections, or if pneumonia is suspected : 500mg twice daily

Pyelonephritis : 250mg twice daily

Uncomplicated gonorrhoea : single dose of 1g

Lyme disease in adults and children over the age of 12 years : 500mg twice daily for 20 days.

Children :

The usual dose is 125mg twice daily or 10mg / kg twice daily to a maximum of 250mg daily.

For otitis media :

Children less than 2 years of age as usual dosage : 125mg twice daily or 10mg / kg twice daily, to maximum of 250mg daily.

Children over 2 years of age : 250mg twice daily or 15mg / kg twice daily, to a maximum of 500mg daily.

There is no clinical trial data available on the use in children under the age of 3 months.

Direction for mixing:

Shake bottle well to loosen granules. Remove cap and Dose-Master™ adapter. Then, add some water to mix with the granules. After that, shake and top up to 50ml mark. Place the Dose-Master™ adapter and cap. Shake the bottle well to mix the medicine properly.

CONTRAINDICATIONS:

Patients with known hypersensitivity to cephalosporin antibiotics.

PRECAUTIONS:

Special care is indicated in patients who have experienced an allergic reaction to penicillins or other beta-lactams. As with other antibiotics, prolonged use of cefuroxime axetil may result in the overgrowth of non-susceptible organisms (e.g. *Candida*, *Enterococci*, *Clostridium difficile*), which may require interruption of treatment. Pseudomembranous colitis has been reported with the use of broad-spectrum antibiotics; therefore, it is important to consider its diagnosis in patients who develop serious diarrhea during or after antibiotic use.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Axcel Cefuroxime – 125 For Oral Suspension (125mg/5ml), careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reactions occurs, Axcel Cefuroxime – 125 For Oral Suspension (125mg/5ml) must be discontinued immediately and appropriate alternative therapy instituted.

Use in Pregnancy and Lactation :

There is no experimental evidence of embryopathy or teratogenic effects attributable to cefuroxime axetil but, as with all drugs, it should be administered with caution during the early months of pregnancy. Cefuroxime is excreted in human milk, and consequently caution should be exercised when cefuroxime axetil is administered to a nursing mother.

Effects on Ability to Drive and Use Machines :

As this medicine may cause dizziness, patients should be warned to be cautious when driving or operating machinery.

SIDE EFFECTS:

Adverse reactions to cefuroxime axetil have been generally mild and transient in nature. As with other cephalosporins, there have been rare reports of erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (exanthematic necrolysis) and hypersensitivity reactions including skin rashes, urticaria, pruritus, drug fever, serum sickness, and very rarely anaphylaxis. A small proportion of patients receiving cefuroxime axetil have experienced gastrointestinal disturbances, including diarrhea, nausea and vomiting. As with other broad-spectrum antibiotics, there have been reports of pseudomembranous colitis. Headache has also been reported. Eosinophilia and transient increases of hepatic enzyme levels, [ALT (SGPT), AST (SGOT) and LDH] have been noted during cefuroxime axetil therapy. There have been rare reports of thrombocytopenia and leucopenia (sometimes profound). As with other cephalosporins, jaundice has been reported very rarely. Cephalosporins as a class tend to be absorbed onto the surface of red cells membranes and react with antibodies directed against the drug to produce a positive Coombs' test (which can interfere with cross-matching of blood) and very rarely haemolytic anaemia.

DRUG INTERACTIONS:

Drugs which reduce gastric acidity may result in a lower bioavailability of this antibiotic compared with that of the fasting state and tend to cancel the effect of enhanced absorption after food. As a false negative result may occur in the ferricyanide test, it is recommended that either the glucose oxidase or hexokinase methods are used to determine blood/plasma glucose levels in patients receiving cefuroxime axetil. This antibiotic does not interfere in the alkaline picrate assay for creatinine.

OVERDOSAGE AND TREATMENT:

Overdosage of cephalosporins can cause cerebral irritation leading to convulsions. Serum levels of cefuroxime can be reduced by haemodialysis and peritoneal dialysis.

STORAGE:

Keep container well closed. Store below 30°C. Protect from light. After reconstitution, refrigerate between 2°C - 8°C and use within 10 days.

KEEP OUT OF REACH OF CHILDREN.

JAUHI DARI KANAK-KANAK.

PACK QUANTITIES :

Available in amber glass bottle of 50ml.

Further information can be obtained from pharmacist, physician and the manufacturer.

Manufactured By & Product Registration Holder :

Kotra Pharma (M) Sdn. Bhd.
 No. 1, 2 & 3, Jalan TTC 12, (90082-V)
 Cheng Industrial Estate,
 75250 Melaka, Malaysia.

MAL14095051AZ

Regn - 111121
 Date of revision: 11/11/21